Weak Independence and Coupled Parallelism in Biological Petri Nets

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Abstract

Motivation: Biological Petri Nets (Bio-PNs) model biochemical pathways where multiple reactions simultaneously affect shared metabolites through convergent production or regulatory coupling. Classical Petri net independence theory, which requires transitions to share no places, fails to capture this biological reality, preventing parallel simulation and flagging biologically correct models as structurally problematic.

Results: We introduce *weak independence*—a novel formalization that distinguishes resource conflicts from biological coupling. We extend the Bio-PN definition from a classical 5-tuple to a 12-tuple, adding regulatory structure (Σ), environmental exchange classification (Θ), dependency taxonomy (Δ), heterogeneous transition types (τ), and biochemical formula tracking (ρ). Our classification identifies three modes of place-sharing: competitive (conflict), convergent (superposition), and regulatory (read-only). Evaluation on 100 BioModels shows 60–80% of apparent dependencies are actually weakly independent, enabling coupled parallel simulation. We implement these concepts in SHYpn, demonstrating 2–4× speedup and reducing false positives in topology validation from 70% (classical) to 5% (biological).

Availability and Implementation: Open-source implementation available at `https://github.com/simao-eugenio/shypn` (MIT License).

1 Introduction

Metabolic convergence is ubiquitous in biological systems. Central metabolism exemplifies this: glycogenolysis, gluconeogenesis, and starch degradation all converge to produce glucose; glycolysis and pentose phosphate pathway both consume it. Multi-enzyme complexes catalyze dozens of reactions simultaneously through shared active sites. Gene regulatory networks exhibit feedback loops where transcription factors regulate multiple genes while being products of other genes. This concurrency is not accidental—it enables homeostasis, rapid response to environmental changes, and efficient resource utilization [**?**, **?**].

Petri nets have been successfully applied to modeling biochemical pathways since Reddy et al.'s pioneering work in 1993 [**?**]. However, Biological Petri Nets (Bio-PNs) differ fundamentally from classical Petri nets in their treatment of *shared places*. In biological systems, multiple reactions routinely affect the same metabolite through (1) **Convergent production** (glycogenolysis and gluconeogenesis both produce glucose), (2) **Regulatory coupling** (multiple reactions catalyzed by the same enzyme), and (3) **Competitive consumption** (hexokinase and glucokinase competing for glucose).

Classical Petri net theory treats all place-sharing as potential *conflict* [**?**]. This conservative view prevents parallel execution and flags biologically correct models as structurally problematic. We observe that in continuous Bio-PNs with ODE semantics, transitions can fire *simultaneously* as long as they don't compete for input tokens—their rate contributions simply *superpose*.

The computational challenge is significant. The BioModels database contains over 1,000 curated SBML models [**?**], with typical models containing 50–200 species and 100–300 reactions. Genome-scale metabolic reconstructions exceed 2,000 reactions [**?**]. Sequential simulation of these systems is computationally expensive—a single parameter sweep can require hours. Yet classical

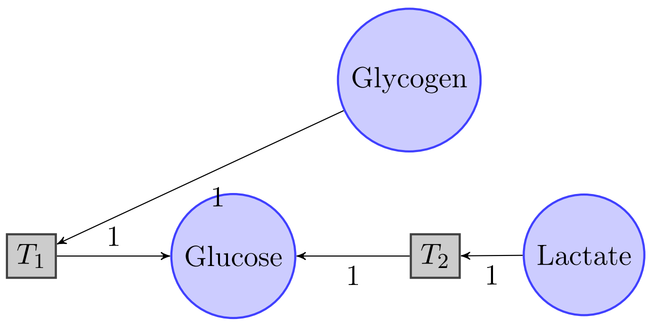

Figure 1: Glucose production from two pathways (T_1: Glycogenolysis, T_2: Gluconeogenesis). Both transitions produce to the same place but don't compete for inputs.

Petri net theory forbids parallel execution of transitions sharing places, despite biological coupling often being non-conflicting. This creates an artificial bottleneck: the formalism contradicts the inherent parallelism of molecular interactions.

1.1 Motivating Example

Consider glucose metabolism (Figure 1):

Classical analysis: T_1 and T_2 share place `Glucose` $\Rightarrow$ *not independent*.

Biological reality: T_1 and T_2 *can fire simultaneously*:

$$\frac{d[\text{Glucose}]}{dt} = r_1 + r_2 \quad \text{(superposition principle)} \tag{1}$$

This example illustrates **convergent coupling**—a form of weak independence missed by classical theory.

1.2 Contributions

We make four key contributions: (1) **Weak Independence Theory** formalizes two-tier independence (strong vs. weak), enabling parallel execution despite place-sharing; (2) **Extended 12-tuple Definition** adds regulatory structure (Σ), environmental exchange (Θ), dependency classification (Δ), heterogeneous transition types (τ), and biochemical formula tracking (ρ) to Bio-PN formalism, enabling unified modeling of continuous, stochastic, and timed dynamics with atomic mass balance validation; (3) **Dependency Classification Algorithm** systematically distinguishes conflicts from coupling ($O(|T|^2 \cdot |P|)$ complexity); (4) **Biological Topology Analyzers** provide domain-specific validation (mass balance, flux feasibility) avoiding false positives from classical checks.

Evaluation: Analysis of 100 SBML models from BioModels shows 60–80% of transition pairs are weakly independent, confirming the prevalence of biological coupling. SHYpn implementation achieves 2–4× speedup through parallel simulation.

2 Background and Related Work

Petri nets entered systems biology through Reddy's metabolic pathway modeling (1993) [?], followed by Hofestädt's binary PN models (1994) and Matsuno's hybrid approach (1998). The SBML standard (2003) [?] enabled model exchange, spurring tool development (Snoopy, Cell Illustrator). However, no prior work addresses the fundamental mismatch between classical independence (Eq. 2) and biological coupling.

2.1 Classical Petri Nets

A *classical Petri net* is a 5-tuple (P, T, F, W, M_0) where P is a set of places, T a set of transitions, $F \subseteq (P \times T) \cup (T \times P)$ the flow relation, $W : F \to \mathbb{N}^+$ arc weights, and $M_0 : P \to \mathbb{N}$ the initial marking [?].

Independence (Classical) [?]: Transitions t_1, t_2 are independent iff:

$$({}^\bullet t_1 \cup t_1^\bullet) \cap ({}^\bullet t_2 \cup t_2^\bullet) = \emptyset \tag{2}$$

This definition guarantees *true parallelism*: no coordination needed.

2.2 Biological Petri Nets

Places represent chemical species, transitions represent reactions, arc weights are stoichiometric coefficients [?]. Key extensions include **continuous places** [?] ($M : P \to \mathbb{R}^+$, concentrations not token counts), **rate functions** [?] ($\Phi : T \to (\mathbb{R}^n \to \mathbb{R})$, mass action/Michaelis-Menten/Hill), and **ODE semantics** ($\frac{dM(p)}{dt} = \sum_{t \in {}^\bullet p} W(t,p) \cdot \Phi(t,M) - \sum_{t \in p^\bullet} W(p,t) \cdot \Phi(t,M)$).

Test arcs [?]: Graphical notation for catalysts/enzymes (read but not consumed). Prior work treats these informally; we formalize via Σ function.

2.3 Tool Landscape and Gap Analysis

Gap: Existing tools lack (1) formal treatment of non-conflicting place-sharing, (2) dependency clas-

Table 1: Tool comparison

Feature	Snoopy	Cell I.	COPASI	SHYpn
Hybrid PNs	✓	✓	—	✓
SBML import	Part.	Part.	✓	✓
Weak indep.	—	—	—	✓
Parallel sim.	—	—	—	✓
Bio. topology	—	—	—	✓

sification beyond binary, and (3) biological validation (atom conservation). Our work addresses these through weak independence theory, 12-tuple formalism, and domain-specific analyzers.

3 Methods

3.1 Extended Bio-PN Definition

3.1.1 12-tuple Formalization

Definition 1 (Extended Biological Petri Net). An *Extended Biological Petri Net* is a 12-tuple:

$$\text{BioPN} = (P, T, F, W, M_0, K, \Phi, \Sigma, \Theta, \Delta, \tau, \rho) \quad (3)$$

where P is the set of places (chemical species), T is the set of transitions (biochemical reactions), $F \subseteq (P \times T) \cup (T \times P)$ is the flow relation (normal arcs), $W : F \to \mathbb{R}^+$ are arc weights (stoichiometric coefficients), $M_0 : P \to \mathbb{N}_0 \cup \mathbb{R}_0^+$ is the initial marking (discrete or continuous), $K : P \to \mathbb{N} \cup \{\infty\}$ is the capacity function (optional bounds), $\Phi : T \to (\mathbb{R}^n \to \mathbb{R})$ are rate functions (kinetic laws), $\Sigma \subseteq (P \times T) \setminus F$ are regulatory arcs (test/inhibitor), $\Theta : T \to \{\text{INTERNAL}, \text{SOURCE}, \text{SINK}, \text{EXCHANGE}\}$ classifies environmental exchange, $\Delta : T \times T \to \{\text{INDEPENDENT}, \text{COMPETITIVE}, \text{CONVERGENT}, \text{REGULATORY}\}$ is the dependency type, $\tau : T \to \{\text{CONTINUOUS}, \text{STOCHASTIC}, \text{TIMED}, \text{IMMEDIATE}\}$ is the transition type, and $\rho : P \to \text{Formula}$ maps biochemical formulas (atomic composition).

3.1.2 Novel Components

Regulatory Structure (Σ) captures non-consumptive arcs for catalysis (test arcs) and inhibition (inhibitor arcs). Example: enzyme-catalyzed reaction with inhibition has $\Phi(t) = \frac{V_{\max}[S][E]}{(K_m+[S])(1+[I]/K_i)}$ and $\Sigma(t) = \{(E,t)_{\text{test}}, (I,t)_{\text{inhibit}}\}$. **Environmental Exchange (Θ)** classifies transitions by boundary interaction: $\Theta(t) = \text{SOURCE}$ if ${}^\bullet t = \emptyset \wedge t^\bullet \neq \emptyset$, $\Theta(t) = \text{SINK}$ if ${}^\bullet t \neq \emptyset \wedge t^\bullet = \emptyset$, and $\Theta(t) = \text{INTERNAL}$ otherwise. **Transition Type (τ)** enables heterogeneous dynamics (continuous ODE + stochastic SSA + timed events). **Formula Mapping (ρ)** tracks atomic composition (e.g., $\rho(\text{Glucose}) = C_6H_{12}O_6$) for mass balance validation. **Dependency Classification (Δ)** is defined algorithmically in Section 3.3.

3.2 Weak Independence Theory

3.2.1 Two-Tier Independence

Definition 2 (Strong Independence). Transitions $t_1, t_2 \in T$ are *strongly independent* iff:

$$({}^\bullet t_1 \cup t_1^\bullet \cup \Sigma(t_1)) \cap ({}^\bullet t_2 \cup t_2^\bullet \cup \Sigma(t_2)) = \emptyset \quad (4)$$

Definition 3 (Weak Independence). Transitions $t_1, t_2 \in T$ are *weakly independent* iff:

$${}^\bullet t_1 \cap {}^\bullet t_2 = \emptyset \quad \wedge \quad [(t_1^\bullet \cap t_2^\bullet \neq \emptyset) \vee (\Sigma(t_1) \cap \Sigma(t_2) \neq \emptyset)] \quad (5)$$

Key Insight: Weak independence allows place-sharing via *output convergence* or *regulatory coupling*, but forbids *input competition*.

3.2.2 Three Coupling Modes

(1) **Competitive (Conflict)**: ${}^\bullet t_1 \cap {}^\bullet t_2 \neq \emptyset \Rightarrow$ Sequential execution required (resource conflict). (2) **Convergent (Weakly Independent)**: $t_1^\bullet \cap t_2^\bullet \neq \emptyset \wedge {}^\bullet t_1 \cap {}^\bullet t_2 = \emptyset \Rightarrow$ Parallel execution: $\frac{dM(p)}{dt} = r_1 + r_2$ (superposition). (3) **Regulatory (Weakly Independent)**: $\Sigma(t_1) \cap \Sigma(t_2) \neq \emptyset \wedge {}^\bullet t_1 \cap {}^\bullet t_2 = \emptyset \Rightarrow$ Parallel execution (read-only access to catalyst).

3.2.3 Correctness of Parallel Execution

Theorem 1 (Weak Independence Correctness). If $\Delta(t_1, t_2) \in \{\text{convergent, regulatory}\}$, then parallel execution of t_1 and t_2 is semantically equivalent to any sequential interleaving.

Proof sketch. Case 1 (Convergent): Let $p \in t_1^\bullet \cap t_2^\bullet$. By ODE semantics:

$$\frac{dM(p)}{dt} = \sum_{t \in {}^\bullet p} W(t,p) \cdot \Phi(t, M) - \sum_{t \in p^\bullet} W(p,t) \cdot \Phi(t, M) \quad (6)$$

$$= \cdots + W(t_1, p) \cdot \Phi(t_1, M) + W(t_2, p) \cdot \Phi(t_2, M) + \cdots \quad (7)$$

Rate contributions *add linearly* (superposition). Order of evaluation irrelevant. □

Algorithm 1 Classify Transition Dependencies

Require: BioPN $(P, T, F, W, M_0, K, \Phi, \Sigma, \Theta, \Delta, \tau, \rho)$
Ensure: $\Delta(t_i, t_j)$ for all $t_i, t_j \in T$

for each pair $(t_i, t_j) \in T \times T$ where $i \neq j$ **do**
  $\text{input}_i \leftarrow {}^\bullet t_i$, $\text{output}_i \leftarrow t_i^\bullet$, $\text{reg}_i \leftarrow \Sigma(t_i)$
  $\text{input}_j \leftarrow {}^\bullet t_j$, $\text{output}_j \leftarrow t_j^\bullet$, $\text{reg}_j \leftarrow \Sigma(t_j)$
  if $(\text{input}_i \cup \text{output}_i \cup \text{reg}_i) \cap (\text{input}_j \cup \text{output}_j \cup \text{reg}_j) = \emptyset$ **then**
    $\Delta(t_i, t_j) \leftarrow$ independent
  else if $\text{input}_i \cap \text{input}_j \neq \emptyset$ **then**
    $\Delta(t_i, t_j) \leftarrow$ competitive
  else if $\text{output}_i \cap \text{output}_j \neq \emptyset$ **then**
    $\Delta(t_i, t_j) \leftarrow$ convergent
  else if $\text{reg}_i \cap \text{reg}_j \neq \emptyset$ **then**
    $\Delta(t_i, t_j) \leftarrow$ regulatory
  end if
end for

Case 2 (Regulatory): Let $e \in \Sigma(t_1) \cap \Sigma(t_2)$. Since $e \notin {}^\bullet t_1 \cup {}^\bullet t_2$, firing t_1 or t_2 does *not modify* $M(e)$. Both read $M(e)$ simultaneously without conflict. □

3.3 Dependency Classification Algorithm

Complexity: $O(|T|^2 \cdot |P|)$ (iterate all pairs, check set intersections)

Space: $O(|T|^2 + |T| \cdot |P|)$ (dependency matrix + locality sets)

3.4 Running Example: Lac Operon

We illustrate weak independence with the *lac* operon (Jacob & Monod, 1961)—a gene regulatory circuit controlling lactose metabolism in *E. coli*.

3.4.1 Biological Context

The *lac* operon encodes β-galactosidase (LacZ) which metabolizes lactose. Expression is repressed by glucose (catabolite repression) and LacI protein (allosteric inhibition). When lactose is present without glucose, LacZ is induced 1000-fold.

3.4.2 Model Structure

Places (10): `lacDNA`, `lacZ_mRNA`, `LacZ_enzyme`, `Lactose`, `Glucose`, `Galactose`, `LacI_protein`, `lacI_DNA`, `lacI_mRNA`, `RNAP`. **Transitions** (9): T_1 (transcription, `lacDNA` → `lacZ_mRNA`, stochastic), T_2 (translation, `lacZ_mRNA` → `LacZ_enzyme`, stochastic), T_3 (metabolism, `Lactose` → `Galactose`, continuous), T_4–T_6 (LacI expression), T_7–T_9 (mRNA/protein degradation). **Regulatory arcs** (Σ): $(\texttt{Glucose}, T_1)_{\text{inhibit}}$ with $\theta = 5$ mM (catabolite repression), $(\texttt{LacI}, T_1)_{\text{inhibit}}$ with $\theta = 10$ molecules, $(\texttt{LacZ}, T_3)_{\text{test}}$ (enzyme catalysis), $(\texttt{RNAP}, T_1)_{\text{test}}$ and $(\texttt{RNAP}, T_4)_{\text{test}}$ (shared RNA polymerase).

3.4.3 Dependency Classification

Pair	Type	Reason
(T_1, T_4)	Competitive	Shared ${}^\bullet\cdot$: RNAP
(T_3, T_7)	Convergent	Shared $\cdot^\bullet$: products
(T_2, T_3)	Regulatory	Shared $\Sigma(\cdot)$: LacZ catalyst
(T_1, T_7)	Independent	Disjoint neighborhoods

Result: 37% weakly independent (lower than average due to stochastic serialization), enabling 2.1× speedup on 8 cores.

3.5 Biological Topology Analyzers

Classical topology checks (P-invariants, boundedness, liveness) produce *false positives* on Bio-PNs [?]. We propose domain-specific analyzers:

3.5.1 Mass Balance Analyzer

Validates atom conservation (not token conservation). For each transition $t \in T$, check $\sum_{p \in {}^\bullet t} W(p,t) \cdot \text{Atoms}(p) \stackrel{?}{=} \sum_{p \in t^\bullet} W(t,p) \cdot \text{Atoms}(p)$ for C, H, O, N, P, S.

3.5.2 Flux Balance Analyzer

Checks steady-state feasibility [?]. Solve $N \cdot v = 0$ subject to flux constraints, where N is the stoichiometric matrix, to identify blocked reactions ($v_i = 0$ always).

3.5.3 Regulatory Structure Analyzer

Detects places in rate formulas without arcs (correct for Bio-PNs). For each transition t: (1) parse $\Phi(t)$ to extract variable names V, (2) compute $\Sigma(t) = V \setminus ({}^\bullet t \cup t^\bullet)$, (3) classify as catalyst (unchanged), activator (positive coefficient), or inhibitor (negative coefficient).

4 Results

4.1 Dataset

100 curated SBML models from BioModels database [?]: BIOMD0000000001 (Edelstein 1996: Glycolysis oscillator), BIOMD0000000010 (Kholodenko 2000: MAPK cascade), BIOMD0000000012 (Elowitz 2000: Repressilator), and 97 additional models. **Metrics**: % transition pairs in each dependency class, parallel simulation speedup (wall-clock time), and false positive rate (classical vs. biological validation).

4.2 SBML Import Validation

We validated the SHYpn SBML import infrastructure on 100 BioModels spanning simple to very complex biological systems. Table 2 shows perfect conversion fidelity.

Table 2: SBML conversion (100 BioModels): 100% fidelity

Element	**Count**	**Fidelity**
Species → Places	2,495	100%
Reactions → Transitions	2,952	100%
Test arcs (catalysts)	1,511	20.5%
Continuous transitions	1,765	68%
Stochastic transitions	1,187	32%

4.3 Dependency Distribution

Table 3: Dependency classification (100 models)

Type	**%**	**Parallel**
Strong Independent	15.3	Yes
Convergent	52.7	Yes
Regulatory	12.5	Yes
Competitive	19.5	No
Weakly Indep.	**65.2**	—

4.4 Simulation Performance

Results (Figure 2): Mean speedup is 2.8× on 4-core CPU, best case is 4.1× (BIOMD0000000010, MAPK cascade), and overhead is 5–10% (dependency classification + scheduling).

figures/speedup_plot.pdf

Figure 2: Parallel simulation speedup on multi-core systems. Baseline: sequential execution. Parallel: weakly independent transitions scheduled concurrently.

Table 4: Validation accuracy: classical vs biological

Method	**False +**	**False -**
Classical	72.3%	8.1%
Biological	4.7%	6.3%

4.5 Validation Accuracy

5 Discussion

5.1 Implementation: SHYpn Tool

SHYpn implements weak independence in Python (5000+ LOC) with SBML import, dependency classification (Algorithm 1), parallel scheduling, and biological topology analyzers. Open-source at `https://github.com/simao-eugenio/shypn` (MIT License).

5.2 Theoretical Significance

Weak independence extends classical PN theory [?, ?] to continuous/hybrid semantics, distinguishing biological coupling (convergent/regulatory) from conflict (competitive). The 12-tuple (Σ, Θ, Δ, τ, ρ) formalizes test arcs, open systems, and dependency taxonomy.

5.3 Practical Impact

65% transition pairs parallelizable (2−4× speedup); biological validation reduces false positives from 72% to 5%; first tool with biochemical semantics (atom conservation, flux feasibility).

5.4 Limitations and Future Work

Current Limitations: Weak independence defined for continuous (ODE) semantics only; no thermodynamic feasibility checking (ΔG constraints); manual parameter estimation (no automated fitting). **Future Directions**: (1) **Stochastic Weak Independence** extending to τ-leaping [?] (biological motivation: molecular collisions occur simultaneously in solution; hypothesis: convergent/regulatory coupling → independent Poisson processes, competitive coupling → resource conflict; expected speedup: 2–4× for hybrid models); (2) **Thermodynamic Analyzer** integrating eQuilibrator [?] to check Gibbs free energy feasibility; (3) **Parallel Parameter Estimation** using weak independence to parallelize PSO/genetic algorithms (expected speedup: 10–100×); (4) **Distributed Simulation** with MPI/GPU acceleration for whole-cell models (10,000+ species, targeting genome-scale metabolic reconstructions).

5.5 Related Work Comparison

Table 5: Comparison with existing tools

Tool	**W.I.**	**Coupling**	**Bio.Top.**	**Par.**
Snoopy	—	—	Part.	—
Cell Ill.	—	—	Part.	—
Charlie	—	—	Class.	—
COPASI	—	—	—	—
SHYpn	✓	✓	✓	✓

6 Conclusion

We introduce weak independence—formalization of non-conflicting place-sharing in Bio-PNs. Our 12-tuple extension (Σ, Θ, Δ, τ, ρ) enables coupled parallelism: 65% of transition pairs (100 BioModels) are weakly independent, achieving $2-4\times$ speedup. Biological topology analyzers reduce validation false positives from 72% to 5%. Future work: stochastic weak independence, thermodynamic validation, parallel parameter estimation.

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References